Prog7313

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POE Part 1

Group 15

Planning and Design

1. Introduction

This section provides an overview of the budget tracker app's conceptual framework while describing its features and user interface design along with its development plan. Research findings serve as the foundation for the design to create a user-friendly and efficient financial management tool.

2. Overview of the App

- **App Name:** BudgetBuddy
- **App Icon:**



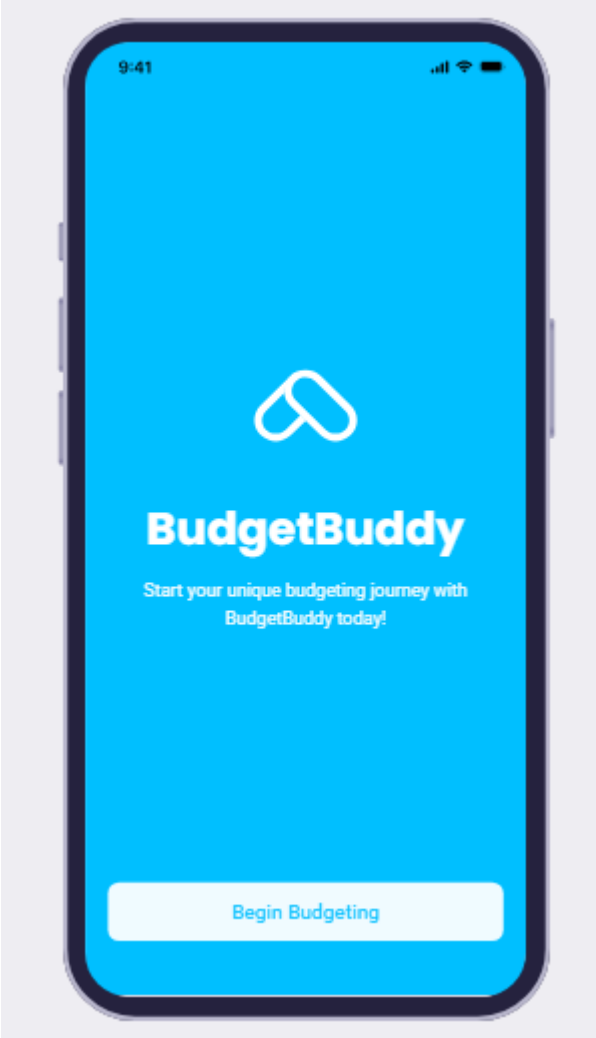
- **Innovative Features:**
    - AI-based expense categorisation
    - Gamification elements
    - Shared budgeting and financial goal setting
    - Voice-activated commands for quick expense logging
    - Interactive financial coaching tips

3. Detailed List of Requirements

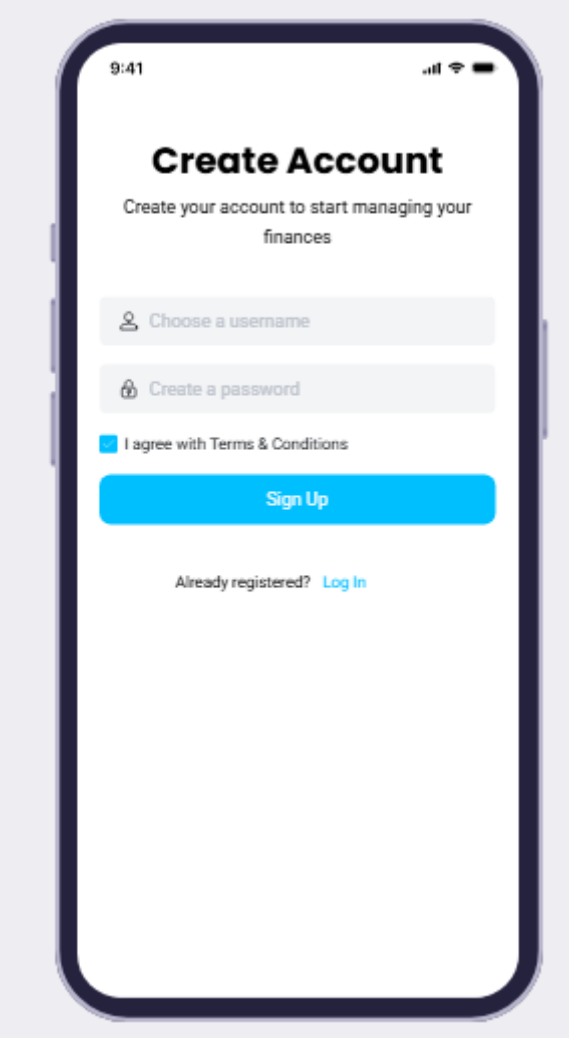
1. **User Registration & Authentication** - Secure login via username and password.
2. **Expense Categorisation** - Customisable categories such as Groceries, Entertainment, etc.
3. **Manual & Automated Expense Entry** - Users can manually enter expenses or automate via bank integration.
4. **Receipt Photo Attachment** - Users can attach receipts to expenses.
5. **Monthly Budget and Category Limits** - Define and track spending limits.
6. **Expense List View & Filtering Options** - View all expenses and filter by category/date.
7. **Data Storage** - Supports both **local database (SQLite)** and **cloud-based storage**
8. **Graphical Spending Insights** - Visualise spending trends with charts and graphs.
9. **Progress Dashboard** - Displays budget performance with alerts for overspending.
10. **Gamification & Reward System** - Encourages responsible spending with achievements and rewards.

4. User Interface Design

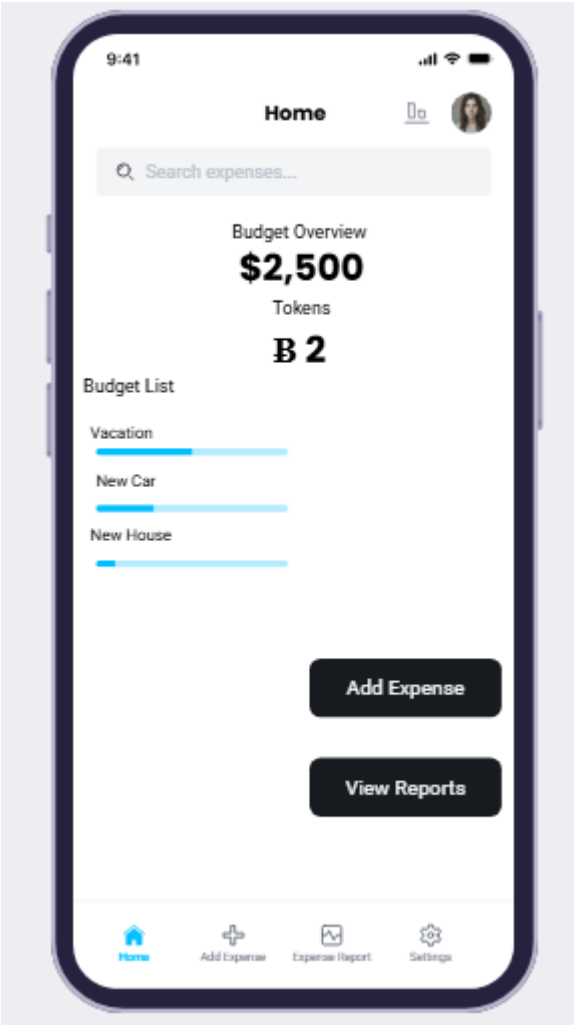
- **Wireframes:**



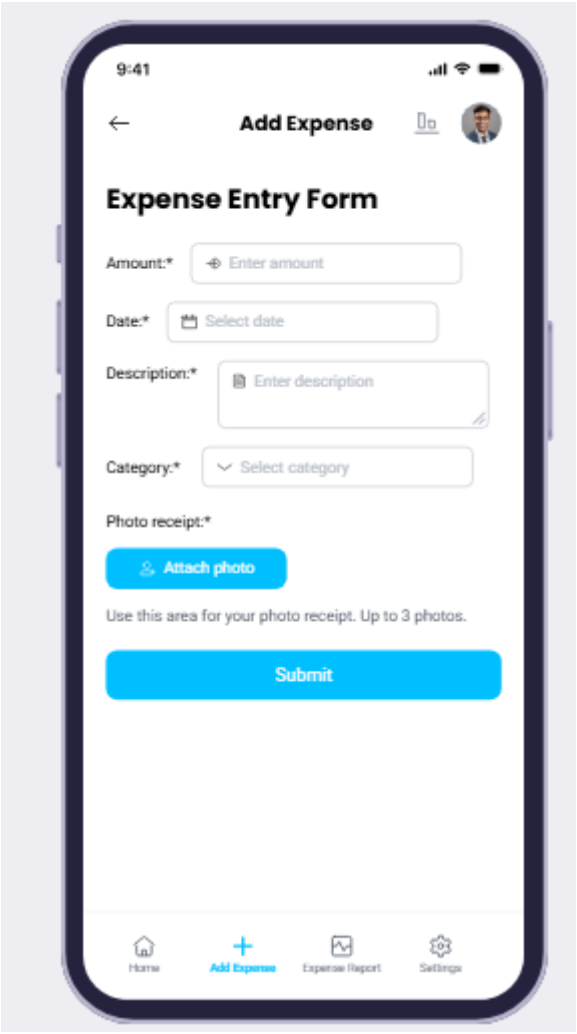
Purpose: Welcomes the user to the app in an inviting way.



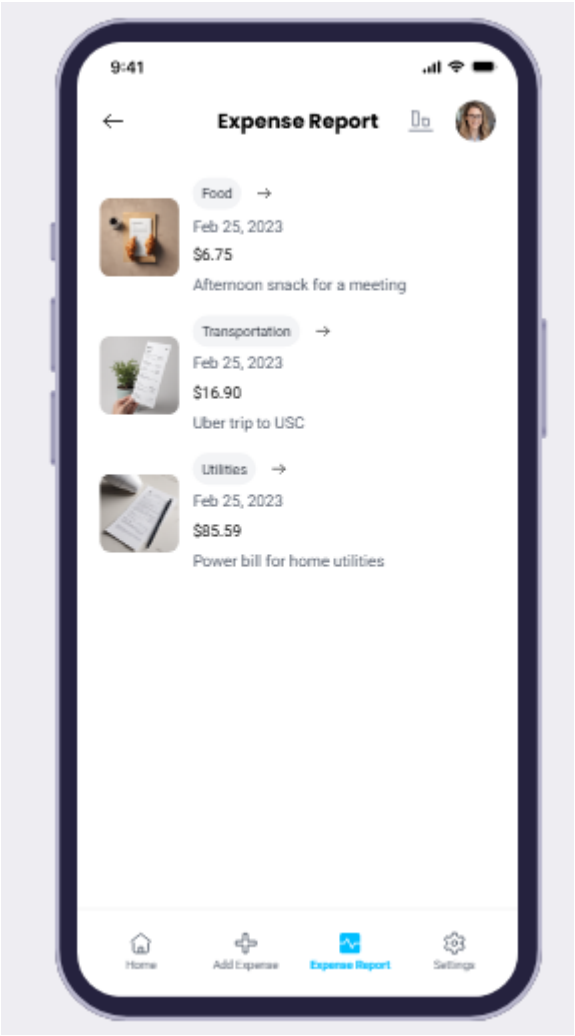
Purpose: Allows the user to create an account with a personal profile to track and manage their budgets.



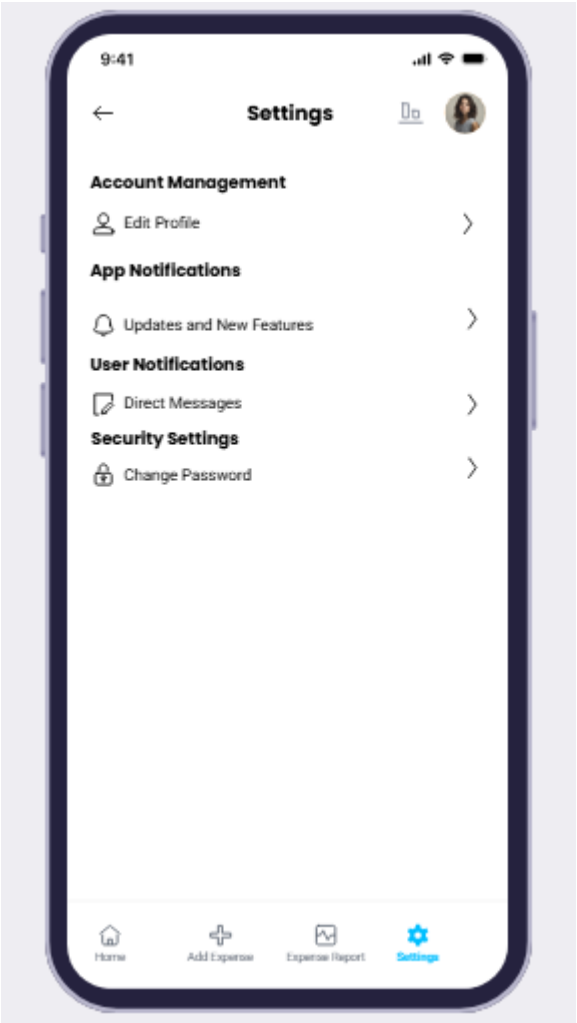
Purpose: An introduction to the app which enables users to conveniently move between all available functions and sections. Users can easily access diverse information through this app which helps them stay engaged because of its convenient viewing features. The application includes a gamified loading bar for each budget and expense report under the budget list that fills progressively as users save and budget effectively to earn BudgetBuddy Tokens (Ƀ) which can be converted to cash or vouchers then used at specific stores (similar to FNB eBucks) with increasing difficulty to fill subsequent bars which creates an engaging and fun user experience.



Purpose: Allows the user to add expenses that they would like to budget for, these will be tracked for the user once they have added the relevant information.

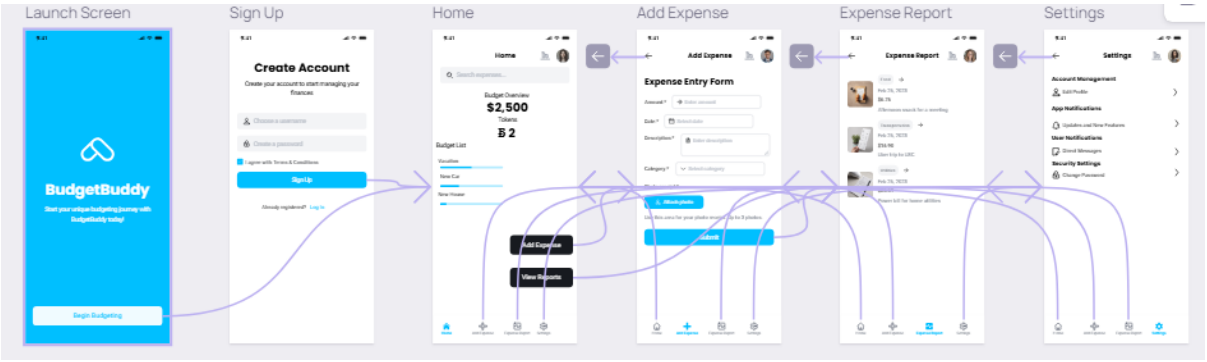


Purpose: The program provides full budget reports for each relevant category which allows users to investigate the details of their budgeting performance. Each report will include charts and graphs to help users easily view and understand their budget tracking.



Purpose: The system enables users to get alerts when their budget status changes including successes in reaching goals and indications of poor budget management. The screen enables users to obtain app updates and modify their profile information for personalised customisation. The system enables users to secure their app by adding a password.

- **Navigation Flow:**



- **Launch Page / screen** → Click Button (Begin Budgeting)

- ***Sign up / login page / screen*** → Create account / login

- **Home Page / screen** → Search for additional expenses (search-bar)
  → Budget overview (Displays budget overview)
  → Token Amount (Based on fillable goal bar - see page description)
  → Budget List (Displays a list as shown)
  → Add expenses (Button)
  → View reports (Button)
  → NavBar (For easy navigation of the app)

- **Add expenses Page / screen** → Expense entry form (Must fill in required fields)
  → Attach photo / receipt (Button)
  → Submit (Button to submit and store the information the user has filled in as well as the attached images)
  → NavBar (For easy navigation of the app)

- **Expense Reports page / screen** → spending trends
  → Budget performance
  → NavBar (For easy navigation of the app)

- **Settings page / screen** → Account management
  → App Notifications
  → User Notifications
  → Security settings
  → NavBar (For easy navigation of the app)

5. Project Plan

Phase 1: Research (Duration: 2 weeks)

Week 1-2

Rayhaan: Research existing budgeting apps for feature comparison.

Ryan: Research technological options for app development (backend and frontend).

Ibrahim: Analyse competitive apps for usability and user experience insights.

Phase 2: Design (Duration: 3 weeks)

Week 3

Rayhaan: Draft initial wireframes for the app's basic layout.

Ibrahim: Start refining wireframes and begin creating interactive prototypes.

Week 4

Ryan: Provide feedback on technical feasibility of design choices.

Rayhaan: Create initial design concepts for the app icon and branding.

Ibrahim: Continue refining prototypes.

Week 5

Ibrahim: Finalise interactive prototypes.

Group: Review and finalise design choices.

Phase 3: Development (Duration: 4 weeks)

Week 6-7

Rayhaan: Develop the frontend of the application, focusing on implementing the designs.

Ryan: Start development of backend services including database management.

Ibrahim: Assist with frontend elements and start developing the documentation.

Week 8-9

Rayhaan: Assist in integrating frontend with backend services.

Ryan: Continue backend development.

Ibrahim: Ensure responsive design and continue documentation development.

Phase 4: Testing and Bug Fixing (Duration: 3 weeks)

Week 10

Rayhaan: Plan and start functional testing.

Ryan: Start performance and security testing.

Ibrahim: Start accessibility testing.

Week 11

Rayhaan: Coordinate necessary adjustments based on testing feedback.

Ryan: Troubleshoot and fix identified bugs.

Ibrahim: Refine user interface based on usability testing feedback.

Week 12

Group: Final testing and fixing of any remaining issues.

Phase 5: Deployment (Duration: 2 weeks)

Week 13

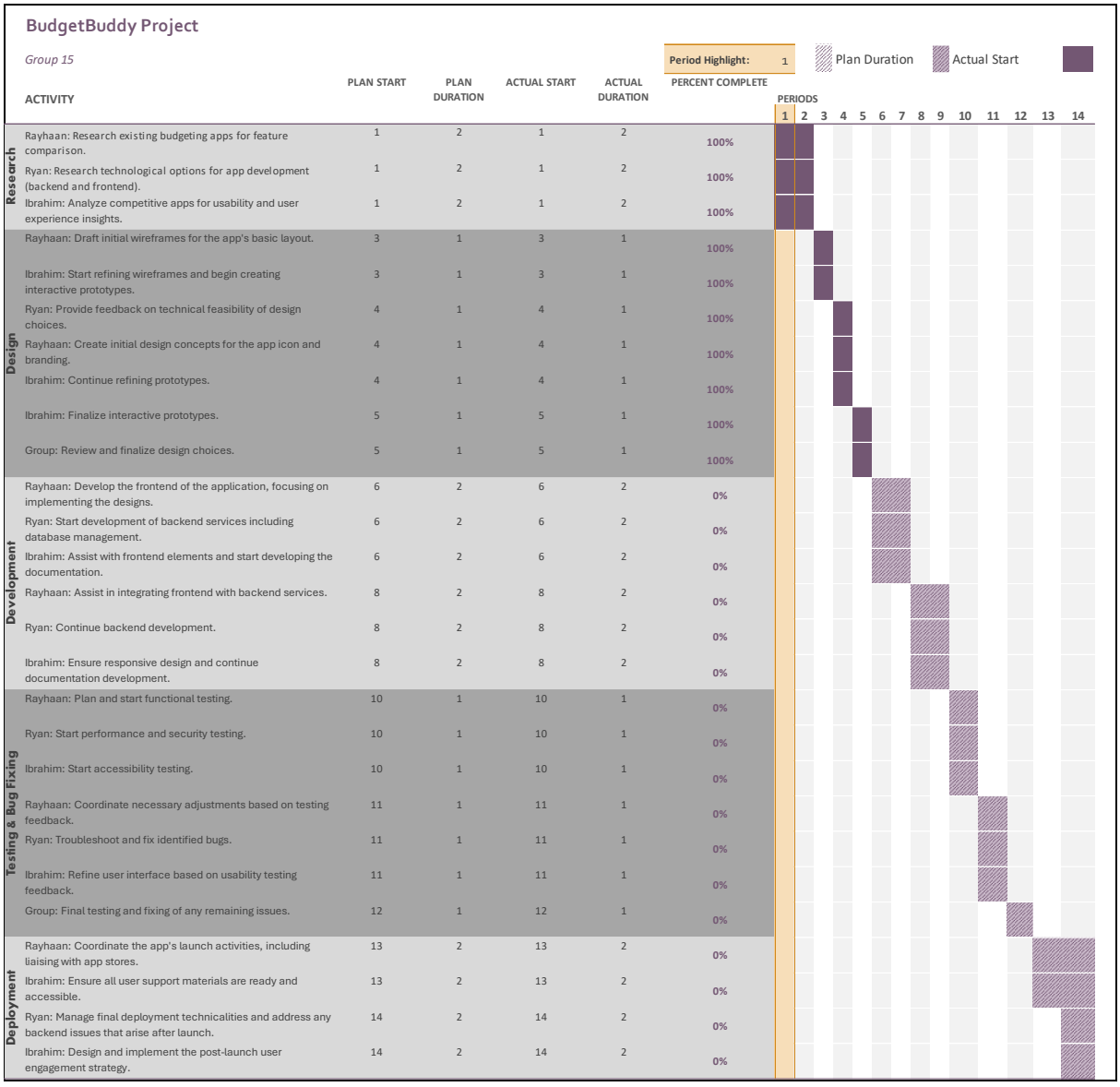
Rayhaan: Coordinate the app's launch activities, including liaising with app stores.

Ibrahim: Ensure all user support materials are ready and accessible.

Week 14

Ryan: Manage final deployment technicalities and address any backend issues that arise after launch.

Ibrahim: Design and implement the post-launch user engagement strategy.



Gantt Chart for BudgetBuddy

6. Conclusion

The document establishes a structured development methodology for BudgetBuddy to create a functional budgeting application that engages users. Through research insights the app design incorporates superior features from current applications and brings in fresh elements to boost user engagement.